Task 11 - Recommend and Defend the Final Design

C Programming Assignment - Rural Electricity Billing System

Data types

Use int64_t for currency in paise and signed intermediate totals, uint64_t for validated non-negative cumulative meter readings, enum for categories and size_t for array sizes. Check arithmetic before adding or multiplying.

Tariff revisions

Load tariff values from a validated configuration file at runtime if revisions must not require recompilation. Validate all rates and effective dates; reject invalid configuration.

Constants

Use #define for compile-time switches and include guards, const for typed read-only objects, and enum for named categories and statuses. Runtime tariffs should not be hard-coded macros.

Evidence

Provide tests for zero usage, each slab boundary, invalid input, rollover, large values and invalid configuration. Compare results with manually calculated bills.

Limitations

Official rates, taxes, subsidies, customer-specific rules and meter-fault policies are not visible in the screenshot. Document assumptions and confirm them with the instructor or co-operative.

District scale

Estimate maximum bill multiplied by maximum household count before selecting types. Use checked arithmetic, audit logs, tariff version tracking and repeatable tests.

Important: illustrative tariff rates are used because the screenshot does not show the complete official tariff table. Replace them with your instructor's prescribed rates.